

Day 29 — Sat 3 Oct

Countdown: 7 days to CS Degree Paper-I

Method: start by recalling yesterday's topic with this page closed. Study the sections below, work every example by hand, then do 20 MCQs.

🎯 Degree — parsing (FIRST/FOLLOW)

Source: ComputerScience_Degree/CSD_02_SystemSoftwareAndCompilers.md

Concept — in plain English first

Application software is software you run *to get a job done that you care about* — a word processor, a browser, a game, a payroll program. **System software** is software that exists *to make other software possible* — it manages the hardware and provides the platform and the tools that applications are written in, translated by, and run on. The user buys the machine for the applications; the system software is the invisible scaffolding underneath.

An analogy: the application is the play the audience came to see; the system software is the theatre, the stage crew, the lighting rig and the translator in the wings.

11.1 Shift-reduce parsing

Four actions on a stack + input:

Action	Meaning
Shift	Push the next input token onto the stack
Reduce	The top of the stack matches a production RHS (a handle); pop it and push the LHS
Accept	Stack = start symbol, input exhausted → success
Error	No valid move

A **handle** is a substring that matches the right side of a production **and** whose reduction is a step in the reverse rightmost derivation. Finding handles is the whole problem.

Two conflicts to name:

- **Shift-reduce conflict:** the parser cannot decide whether to shift or reduce (classic: the dangling-else).
- **Reduce-reduce conflict:** two different productions could be reduced.

Worked shift-reduce parse of `id * id` with $E \rightarrow E+T \mid T$, $T \rightarrow T * F \mid F$, $F \rightarrow id$

Stack	Input	Action
\$	id * id \$	shift
\$ id	* id \$	reduce $F \rightarrow id$
\$ F	* id \$	reduce $T \rightarrow F$
\$ T	* id \$	shift
\$ T *	id \$	shift

\$ T * id	\$	reduce $F \rightarrow id$
\$ T * F	\$	reduce $T \rightarrow T * F$
\$ T	\$	reduce $E \rightarrow T$
\$ E	\$	accept

11.2 The LR family

LR(k) = Left-to-right scan, **R**ightmost derivation in reverse, k lookahead (k=1 in practice). All build a DFA of **items** and a table with **ACTION** (shift/reduce/accept) and **GOTO** (on non-terminals) parts. The four variants trade power for table size:

Parser	Lookahead / states	Power	Table size	Note
LR(0)	none	weakest	small	Reduces without lookahead; conflicts on many grammars
SLR(1)	uses FOLLOW sets	more than LR(0)	small (= LR(0) states)	Simple LR; reduce on a in FOLLOW(A)
CLR / LR(1)	lookahead carried in each item	strongest	largest	Canonical LR; most states
LALR(1)	merges LR(1) states with the same core	between SLR and CLR	small (= LR(0) states)	Used by yacc/bison ; the practical choice

Power ordering: **LR(0) < SLR(1) < LALR(1) < CLR(1) = LR(1)**. Table-size ordering: SLR = LALR = LR(0) states $\ll$ CLR states. LALR is the sweet spot: nearly the power of CLR with the compactness of SLR.

An **LR(0) item** is a production with a dot marking how far parsing has progressed:

$A \rightarrow \alpha \cdot \beta$ means "we have seen α , expect β ". $A \rightarrow \alpha \cdot$ (dot at the end) is a **reduce** item.

Worked LR(0) item sets — augmented grammar

Augment with $S' \rightarrow S$. Grammar:

```
S' → S
S  → C C
C  → c C | d
```

Item-set construction (closure adds items for non-terminals right after the dot; goto moves the dot across a symbol):

```
I0: S'→•S      goto(I0,S)=I1
    S →•CC      goto(I0,C)=I2
    C →•cC      goto(I0,c)=I3
    C →•d       goto(I0,d)=I4

I1: S'→S•      (accept on $)

I2: S →C•C      goto(I2,C)=I5
    C →•cC      goto(I2,c)=I3
    C →•d       goto(I2,d)=I4

I3: C →c•C      goto(I3,C)=I6
    C →•cC      goto(I3,c)=I3
    C →•d       goto(I3,d)=I4

I4: C →d•      (reduce C→d)
```

I5: S →CC• (reduce S→CC)
 I6: C →cC• (reduce C→cC)

SLR(1) table (reduce entries placed under FOLLOW; here FOLLOW(S)={\$}, FOLLOW(C)={c,d,\$}):

State	c	d	\$	S	C
0	s3	s4		1	2
1			acc		
2	s3	s4			5
3	s3	s4			6
4	r(C→d)	r(C→d)	r(C→d)		
5			r(S→CC)		
6	r(C→cC)	r(C→cC)	r(C→cC)		

sN = shift and go to state N; rX = reduce by production X; acc = accept; blank = error. **No cell has a conflict**, so the grammar is SLR(1). Reproducing the item sets I0–I6, the goto arrows, and this table is a full 15-mark answer.

SLR vs LALR vs CLR difference in one line: SLR uses the whole FOLLOW set to decide reductions (crude, can conflict); CLR carries an exact lookahead in every item (precise but many states); LALR merges CLR states with identical cores, keeping the precise-enough lookahead in far fewer states.

🕒 Forensic — scene of crime

Source: ComputerForensic/FOR_01_ForensicScienceFundamentals.md

5.3 Crime scene documentation

The golden rule: document before you disturb. Notes, photographs, video and sketch are **complementary, not alternatives** — a good scene has all four. That sentence alone earns a mark.

(a) Note taking

- **Contemporaneous** — written at the scene as things happen, never reconstructed later.
- Record: date and time of the call, time of arrival, weather and lighting, exact location, who was present, who summoned whom, condition of doors/windows/lights/appliances, position and condition of the body, transient observations (odours, temperature, wetness), description and location of each item of evidence, who collected what and when, time of departure.
- Use a **bound notebook with numbered pages**, ink, no erasures — strike a line through errors and initial them (an erasure suggests alteration).
- Note **negative findings** explicitly ("no forced entry observed at the rear door").
- Notes are **disclosable** and can be demanded in cross-examination — write them as if the defence will read them, because it will.

(b) Photography

Principle: photographs must show the scene **as it was found**, must not distort, and must be

supported by a photograph log.

The required sequence — memorise these three levels:

Level	What it shows	Purpose
1. Overall / long-range	The entire scene and its surroundings, exterior of the building, street, approach routes, all four walls of a room, ceiling and floor	Establishes where the scene is and its general layout — orientation
2. Mid-range / medium	An item of evidence together with a recognisable fixed landmark (a door, a table, a wall)	Establishes the spatial relationship of the evidence to the scene
3. Close-up	The item alone, filling the frame — twice: once WITHOUT a scale, then again WITH a scale and an identification label	Records the detail of the item; the scale permits later measurement and 1:1 comparison

Why "without a scale first"? Because the defence can argue the scale was placed so as to obscure or alter something. The unscaled photograph proves the item's appearance before anything was added to the frame.

Other photography rules

- Photograph **before** any item is moved, marked or numbered — then re-photograph with the evidence markers in place.
- Photograph the scene from **all four corners / all approaches**, and the **ceiling and floor**.
- Take **eye-level** shots to reflect a witness's viewpoint where relevant.
- Photograph the **entrance and exit points** and any signs of forced entry.
- Photograph the **body** from all sides, plus the surface underneath after removal.
- Maintain a **photograph log**: photo number, subject, direction faced, time, camera settings, photographer.
- Do not use filters, do not edit; retain the **original RAW/unaltered files** with hash values.
- **Digital scenes**: photograph the screen, the front and rear of every device, all serial numbers/IMEI/service tags, the cable layout, and the surrounding desk (sticky notes with passwords are real and common).

(c) Videography

- Provides a **continuous, contextual walkthrough** that stills cannot — spatial relationships, scale, and the sense of moving through the scene.
- Start with a **statement to camera**: case number, date, time, location, name of the operator.
- Move **slowly and steadily**, no zoom-happy jerks, no rapid panning.
- **No commentary of opinions** — narration should be purely factual, or the audio may be turned off entirely (a stray speculative remark on the tape is a gift to the defence).
- Do the videography **before** stills and before any evidence markers are placed, and again afterwards if required.
- Do not edit; retain the original file and its hash.
- Videography is now increasingly **mandatory in India** for search and seizure in certain offences under the BNSS 2023 [△](#) verify the section.

(d) Sketching

Why sketch when you have photographs? Because a photograph **distorts perspective and does not give measurements**. A sketch gives **accurate relative distances** and removes irrelevant clutter. Photograph = appearance; sketch = **dimension**.

Type	Made	Contents

Rough sketch	At the scene, freehand, not to scale	All measurements written in, all evidence items numbered, compass north, dimensions of the room, position of doors/windows/furniture/body/evidence. Must never be altered or destroyed — it is the original record and is producible in court.
Finished / final sketch	Later, in the office, drawn to scale (by hand or with CAD)	Neat, to scale, with a legend/key , scale ratio (e.g. 1:50) , north arrow , case number, date, location, name and signature of the preparer, and a note of who took the measurements. Only relevant items retained.

Essential elements of any crime-scene sketch (the "must-have" list):

1. Case number, date and time
2. Location/address
3. **North arrow (compass direction)**
4. **Scale** (or the words "not to scale" on the rough sketch)
5. **Legend/key** identifying numbered items
6. Names of the sketcher and the measurer, with signatures
7. Names of witnesses present

Measurement methods — describe these (very examinable):

Method	How it works	Best for
Rectangular coordinate	Measure the perpendicular distance from an object to two fixed walls at right angles to each other (e.g. 1.2 m from the north wall, 2.4 m from the east wall)	Indoor scenes with square rooms — the standard method
Triangulation	Measure the distance from the object to two (or three) fixed reference points ; the intersection of the arcs fixes the position	Outdoor scenes with no straight walls; irregular areas
Baseline (coordinate)	Establish a fixed straight line (a tape between two fixed points, or a wall); for each item measure the distance along the baseline and the perpendicular offset from it	Large outdoor scenes, roads, open ground
Polar / azimuth coordinate	From a single fixed point, record the distance and the angle (bearing) to each object	Large open areas, aerial/surveying-style work, total-station or 3-D scanner use
Cross-projection / "exploded" sketch	The walls and ceiling are drawn as if folded flat outward from the floor plan, so that evidence on the walls and ceiling (bullet holes, blood spatter) can be plotted with floor evidence on one sheet	Rooms with evidence on vertical surfaces — shootings, blood-spatter scenes
Elevation sketch	A side/vertical view	Height of a bullet hole, a hanging point

Diagram to practise: draw a rectangular room plan with a door, two windows, a body, and three numbered evidence items; annotate two of them with rectangular coordinates and one by triangulation; add a north arrow, scale bar and legend. Do this once a week — it is a ready-made answer to any sketching question.

(e) 3-D scanning technique

Concept: a laser scanner (LIDAR) or structured-light/photogrammetric system placed at several positions in the scene records **millions of measured points** (a "point cloud"), each with X, Y, Z coordinates and often colour. The scans are registered together into a single, dimensionally accurate **3-D digital model of the entire scene**.

How it works:

1. The scanner is placed at a station and rotates through 360°, emitting a laser pulse and timing its return (**time-of-flight**) or measuring **phase shift** to compute distance.

2. Reference **targets/spheres** are placed so that scans from different stations can be **registered** (aligned) into one coordinate system.
3. Multiple stations are scanned to eliminate "shadows" behind objects.
4. Photographs are mapped onto the point cloud for colour/texture.
5. Software produces measurements, floor plans, sections, walkthrough animations and VR views.

Advantages

- **Captures everything, measured, at once** — you can take a measurement months later that nobody thought to take at the scene.
- **Non-contact and non-destructive**; nothing is disturbed.
- **Very fast** compared with manual measuring; a room in minutes.
- Extremely **accurate** (millimetre order) and reduces human measurement error.
- Enables **reconstruction**: bullet trajectories, line-of-sight/visibility analysis, blood spatter point-of-origin, vehicle-collision reconstruction.
- Produces compelling, easily understood **courtroom visualisations**; the scene can be "revisited" virtually after it has been released.
- Permits **virtual walkthroughs** by the judge or jury.

Limitations

- **High cost** of equipment and software; specialised training needed.
- Large **data volumes** and processing time.
- Records **geometry, not micro-detail** — it will not show a latent fingerprint or a fibre; it supplements, never replaces, photography and physical collection.
- **Line-of-sight limited** — occluded areas need extra stations; poor with transparent, glossy or very dark surfaces.
- Weather/vibration sensitive outdoors.
- Requires the court to accept the **validation and authenticity** of the model — and the model is itself electronic evidence requiring a §65B/BSA §63 certificate.